

Rahul Rawat

Mannheim, Germany · rahulrawat2r@gmail.com · 015563603340 · linkedin.com/in/rahulrawat2r/ · github.com/rahulrawat20022002

PROFILE

Master of Science Data Science student at SRH Heidelberg with hands on experience building machine learning solutions on real time and time series data. I have worked from raw data analysis through to productive models, built pipelines that ingest live sensor style feeds, and surfaced results in dashboards for non technical users. I work with Python, scikit learn, gradient boosting libraries and SQL, and I am looking for a working student role in data science and machine learning for intelligent production systems, which is exactly what your Fertigungstechnologien team offers.

EDUCATION

M.Sc. Data Science and Analytics, SRH University of Applied Sciences Heidelberg

Apr 2025 to Present

B.Sc. Computer Science, GL Bajaj Institute of Technology and Management

2019 to 2023

EXPERIENCE

eRay GmbH

Data Scientist · Oct 2025 to Mar 2026

- Built an end to end recursive time series pipeline forecasting four indicators, chlorophyll a, turbidity, pH and dissolved oxygen, taking the work from raw data analysis through to a productive forecasting model.
 - Benchmarked six models head to head, Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost and Prophet, and used CatBoost multi quantile regression to deliver asymmetric 80 percent prediction intervals.
 - Engineered velocity and ecological bounds into the orchestrator that flag physically implausible readings, the same logic that underpins anomaly detection on process data.
 - Reconstructed missing readings with MICE imputation and added gate checks that halt on failed imputation rather than letting bad data cascade.
-

PERSONAL PROJECTS

Real Time Flight Tracking Data Pipeline: Python, PySpark, BigQuery, dbt, Airflow, GCP, Tableau

- Set up real time data collection polling live positions every 30 seconds and enriching each record against four sources, the same kind of real time, sensor style data your IoT platform produces.
- Cleaned raw data with PySpark on Google Cloud, shaped it into analysis ready tables with dbt, and orchestrated the system with Apache Airflow so batch and real time layers refresh automatically every 15 minutes.
- Surfaced findings in a Tableau dashboard, showing measurable effects such as air traffic dropping 4.4 times in heavy rain.

Movie Analytics and Machine Learning Pipeline: Python, SQL, GCP BigQuery, BigQuery ML, Looker Studio

- Built an end to end batch pipeline through a Bronze, Silver and Gold medallion architecture, fully automated on a scheduler with no manual steps.
- Trained a machine learning classifier to predict outcomes before they happen, deliberately splitting features to prevent data leakage.
- Built a five page Looker Studio dashboard for non technical stakeholders, the kind of monitoring view your worldwide production sites would use.

CreditIQ, Fairness by Design Credit Scoring System: Python, scikit learn, AIF360, SHAP

- Built a machine learning pipeline that detects and corrects problematic patterns in a decision process, lifting a fairness ratio from 0.79 to 0.88 and cutting the false negative rate from 44 percent to 16.7 percent.
 - Diagnosed a hidden bias using SHAP, demonstrating the anomaly hunting and root cause analysis this role needs.
 - Backed the pipeline with unit tests at 100 percent branch coverage for reliable, production minded code.
-

RESEARCH AND THESIS

Bachelor Thesis, Diabetes Prediction Using Machine Learning: Python, scikit learn, Pandas, Seaborn

- Built a full end to end machine learning pipeline comparing six classifiers on a clinical dataset of 768 records with 10 fold cross validation and per model confusion matrices.
 - Caught biologically impossible zero values the original authors had overlooked, then applied IQR based outlier removal and proper imputation.
 - Chose ROC AUC over accuracy as the headline metric for a 65 to 35 imbalanced dataset and wrote up the findings as an IEEE style paper with an honest limitations section.
-

LANGUAGES

- English: fluent, professional working proficiency
- German: B1, in progress